

Coimisiún na Scrúduithe Stáit

State Examinations Commission

Junior Cycle 2026

Marking Scheme

Mathematics

Higher Level

Note to teachers and students on the use of published marking schemes

Marking schemes published by the State Examinations Commission are not intended to be standalone documents. They are an essential resource for examiners who receive training in the correct interpretation and application of the scheme. This training involves, among other things, marking samples of student work and discussing the marks awarded, so as to clarify the correct application of the scheme. The work of examiners is subsequently monitored by Advising Examiners to ensure consistent and accurate application of the marking scheme. This process is overseen by the Chief Examiner, usually assisted by a Chief Advising Examiner. The Chief Examiner is the final authority regarding whether or not the marking scheme has been correctly applied to any piece of candidate work.

Marking schemes are working documents. While a draft marking scheme is prepared in advance of the examination, the scheme is not finalised until examiners have applied it to candidates' work and the feedback from all examiners has been collated and considered in light of the full range of responses of candidates, the overall level of difficulty of the examination and the need to maintain consistency in standards from year to year. This published document contains the finalised scheme, as it was applied to all candidates' work.

In the case of marking schemes that include model solutions or answers, it should be noted that these are not intended to be exhaustive. Variations and alternatives may also be acceptable. Examiners must consider all answers on their merits, and will have consulted with their Advising Examiners when in doubt.

Future Marking Schemes

Assumptions about future marking schemes on the basis of past schemes should be avoided. While the underlying assessment principles remain the same, the details of the marking of a particular type of question may change in the context of the contribution of that question to the overall examination in a given year. The Chief Examiner in any given year has the responsibility to determine how best to ensure the fair and accurate assessment of candidates' work and to ensure consistency in the standard of the assessment from year to year. Accordingly, aspects of the structure, detail and application of the marking scheme for a particular examination are subject to change from one year to the next without notice.

General notes on the marking

In general, accept a candidate's work in one part of a question for use in subsequent parts of the question, unless this oversimplifies the work involved.

Scales and Slips
Certain types of mistake – typically involving incorrect rounding, omission of units, a misreading that does not oversimplify the work, or an arithmetical error that does not oversimplify the work – are referred to as a **slip** and are denoted with a *. In such cases, a mark that is one mark below the full-credit mark may be awarded. This level of credit is referred to as *Full Credit –1*. Thus, for example, in Scale 10C, *Full Credit –1* of 9 marks may be awarded.

The only marks that may be awarded for a question are those on the appropriate scale in the marking scheme, or *Full Credit –1*.

Instructions regarding penalties for omitted or incorrect units are given in the scheme for each question to which they apply. A penalty for rounding is applied once per unit of marking (i.e., if parts (a) and (b) are marked as a single unit, a penalty for rounding is only applied once for (a) and (b) combined).

In general, an answer without sufficient supporting work is awarded the lowest level of credit on the scale above *No Credit* (typically *Partial Credit* or *Low Partial Credit*, as appropriate).

Steps
Where steps are listed in the Marking Notes, unless otherwise specified, it is to be taken that they can be independently correct / incorrect – that is, in a candidate's solution, step n can be considered correct even if previous step(s) have not been correctly presented, as long as the work done to arrive at step n from the previous step(s) has not been oversimplified. Note also that these steps may not need to be presented in the order specified in the Marking Notes. In general, it is specifically noted where these do not hold.

Errors
An error is a mistake in a candidate's work that is not considered a slip. Where a question is **not** marked using steps, if a candidate has a single error, they are generally awarded one level of credit below that which they would otherwise have been awarded. Similarly, where they have two errors, they are generally awarded two levels of credit below that which they would otherwise have been awarded. (If they present sufficient work for *Low Partial Credit*, they will be awarded this at a minimum, regardless of the number of errors.) For example, on a C-scale:

- *High Partial Credit*: One error, otherwise fully correct (or fully correct with a *)
- *Low Partial Credit*: Two errors, otherwise fully correct (or fully correct with a *)

Where a question **is** marked using steps, this does not apply. Instead, an error in a step means that the step has not been completed correctly; this does not affect the completion of other steps (unless it oversimplifies the work). So, if a candidate has multiple errors on a single step, they could still be awarded up to *High Partial Credit*, depending on the marking scheme.

Slips and Errors
Where a candidate has a single * on their solution, this is ignored in the awarding of credit unless they would otherwise have *Full Credit*. If they would otherwise have *Full Credit*, then *Full Credit –1* is awarded. Where a candidate has multiple *s, this is generally treated as an error. This applies even where the *s are in different steps.

Multiple answers
Where a solution requires substantial work, all separate attempts are marked and the marks for the best one are awarded, regardless of crossing out.

Where a solution requires selection from the question:

- If a candidate has only one answer, which they have then crossed out, the crossing-out is ignored
- If a candidate has both crossed-out answer(s) and answer(s) that are not crossed out, ignore the crossed-out answer(s) and only mark those that are not crossed out

- If a candidate has multiple answers that are **not** crossed out, award the lowest mark associated with these answers (generally, such a set of answers will end up being considered incorrect).

Excess work
Where a solution requires substantial work, incorrect excess work (i.e. work presented after the answer) is generally not penalised. This does not apply where candidates are asked to write a sentence, or number of sentences, to explain / justify / describe something. In such cases, the answer as a whole is considered.

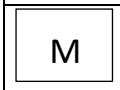
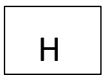
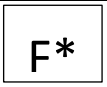
Where a subsequent part of a question requires candidates to use their answer from a previous part, and an incorrect answer arising from incorrect excess work is used, this is generally treated as an error in the subsequent part, unless it oversimplifies the work involved (in which case a lower level of credit is generally awarded for the subsequent part).

Square brackets
Where something is contained in square brackets in the model solution, it is **not** required for *Full Credit*.

Work of merit
Where the scheme indicates "work of merit" (or "substantial work of merit"), examples are given to exemplify the standard of work required to be considered work of merit (or substantial work of merit) in that particular question.

Accept
Where it is indicated to "accept work, this means that, for this year, such work is to be accepted for *Full Credit*, unless otherwise indicated.

Palette of annotations available to examiners

Symbol	Name	Meaning in the body of the work	Meaning when used in the right margin
	Tick	Work of relevance	The work presented in the body of the script merits full credit
	Cross	Incorrect work (distinct from an error)	The work presented in the body of the script merits 0 credit
	Star	Rounding / Unit / Arithmetic error Misreading	
	Horizontal wavy	Error	
	P		The work presented in the body of the script merits a partial credit for B scales
	L		The work presented in the body of the script merits low partial credit
	M		The work presented in the body of the script merits mid partial credit
	H		The work presented in the body of the script merits high partial credit
	F star		The work presented in the body of the script merits Full Credit (– 1)
	Left Bracket		Another version of this solution is presented elsewhere and it merits equal or higher credit
	Vertical wavy	No work on this page (portion of the page)	
WOM		The candidate has presented work or merit	
	Oversimplify	The candidate has oversimplified the work	

Note:	Where work of substance is presented in the body of the script, the annotation on the right margin should reflect a combination of annotations in the work. In a **C scale** that is **not** marked using steps, where * and  and  appear in the body of the work, then  should be placed in the right margin. In the case of a **D scale** with the same annotations,  should be placed in the right margin.
Note:	Where work is presented for a question in the answerspace provided, and a separate attempt is presented on additional pages, both attempts are marked. For each attempt, the annotation for the appropriate credit level is put in the right margin next to that attempt. The annotation for the lower credit level is then left-bracketed, and the mark for the higher credit level is awarded.

Structure of the marking scheme

Candidate responses are marked according to different scales, depending on the types of response anticipated. Scales labelled A divide candidate responses into two categories (correct and incorrect). Scales labelled B divide responses into three categories (correct, partially correct, and incorrect), and so on. The scales and the marks that they generate are summarised in this table:

Scale label	**B**	**C**	**D**
No of categories	3	4	5
5-mark scale	0, 3, 5	0, 2, 3, 5	0, 2, 3, 4, 5
10-mark scale	0, 5, 10	0, 3, 6, 10	0, 3, 5, 7, 10
15-mark scale		0, 5, 10, 15	0, 4, 8, 12, 15
20-mark scale			

A general descriptor of each point on each scale is given below. More specific directions in relation to interpreting the scales in the context of each question are given in the scheme.

Marking scales – level descriptors

A-scales (two categories)

- incorrect response (no credit)
- correct response (full credit)

B-scales (three categories)

- response of no substantial merit (no credit)
- partially correct response (partial credit)
- correct response (full credit)

C-scales (four categories)

- response of no substantial merit (no credit)
- response with some merit (low partial credit)
- almost correct response (high partial credit)
- correct response (full credit)

D-scales (five categories)

- response of no substantial merit (no credit)
- response with some merit (low partial credit)
- response about half-right (mid partial credit)
- almost correct response (high partial credit)
- correct response (full credit)

Model Solutions & Marking Notes

Note: The model solutions for each question are not intended to be exhaustive – there may be other correct solutions. Any Examiner unsure of the validity of the approach adopted by a particular candidate to a particular question should contact his/her Advising Examiner.

<table>
<tr><th>Q1</th><th>Model Solution – 10 marks</th><th>Marking Notes</th></tr>
<tr><td>(a)</td><td>
<table>
<tr><td></td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr>
<tr><td>R</td><td>R1</td><td>R2</td><td>R3</td><td>R4</td><td>R5</td><td>R6</td></tr>
<tr><td>Y</td><td>Y1</td><td>Y2</td><td>Y3</td><td>Y4</td><td>Y5</td><td>Y6</td></tr>
<tr><td>B</td><td>B1</td><td>B2</td><td>B3</td><td>B4</td><td>B5</td><td>B6</td></tr>
</table>
</td><td>Scale 5C (0, 2, 3, 5)
There are 16 entries required.
Note: Accept in reverse order

Low Partial Credit:
• 1 correct entry

High Partial Credit:
• 11 correct entries

Full Credit (−1):
• 15 correct entries</td></tr>
<tr><td>(b)</td><td>(i)
$\frac{6}{18}$ or $\frac{1}{3}$ or $0{\cdot}\dot{3}$

(ii)
$\frac{3}{18}$ or $\frac{1}{6}$ or $0{\cdot}1\dot{6}$

OR

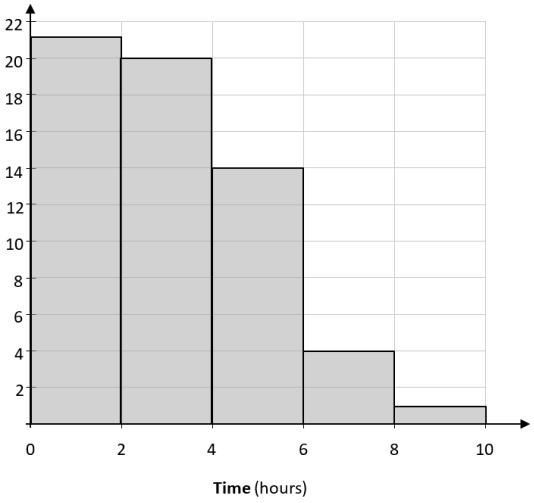
$\frac{1}{3}\times\frac{3}{6}=\frac{3}{18}=\frac{1}{6}$</td><td>10D (0, 3, 5, 7, 10)
Accept correct answers without supporting work

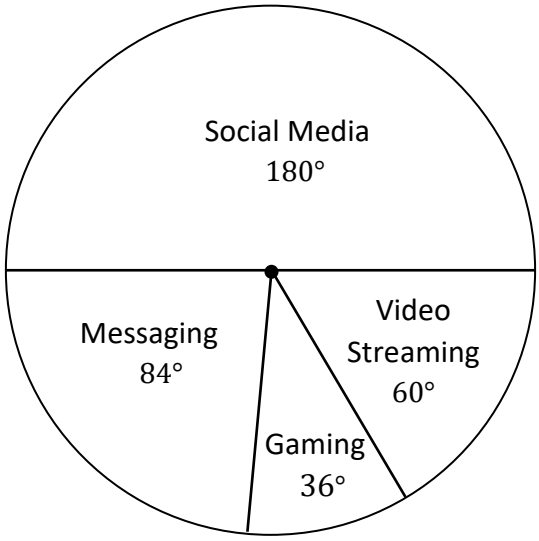
Low Partial Credit:
• Work of merit in one part, for example, a correct numerator or denominator

Mid Partial Credit:
• One part correct
• Work of merit in both parts

High Partial Credit:
• One part correct and work of merit in the other part

Full Credit (-1):
• Answers rounded</td></tr>
</table>

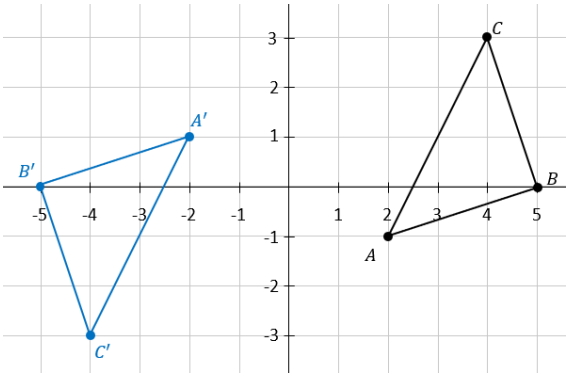
Q2	Model Solution – 25 marks	Marking Notes
(a) (i),(ii)	**(i)**  **(ii)** $21 + 20 \;=\; 41$ [students]	**Scale 10D (0, 3, 5, 7, 10)** Note: Accept 41 without work in **(ii)** *Low Partial Credit:* • Work of merit in one part, for example, in **(i)**, one correct height indicated or bar drawn; in **(ii)**; mentions 21 or 20 *Mid Partial Credit:* • One part correct • Work of merit in both parts *High Partial Credit:* • One part correct and work of merit in the other part
(a)(iii)	$\dfrac{1(21) + 3(20) + 5(14) + 7(4) + 9(1)}{60}$ $= \frac{188}{60} = \frac{47}{15} = 3{\cdot}1333\ldots = 3{\cdot}1$ [hours]	**Scale 5C (0, 2, 3, 5)** Accept correct answer without work *Low Partial Credit:* • Work of merit, for example, indicates division by 60; one correct mid-interval value; numerator with consistent incorrect mid-interval values. *High Partial Credit:* • First line of solution • Consistent incorrect mid-intervals and finishes correctly *Full Credit -1:* • No rounding or incorrect rounding

Q2	Model Solution – 25 marks	Marking Notes
(b)	Social Media: $\frac{30}{60} \times 360 = 180°$ Video Streaming: $\frac{10}{60} \times 360 = 60°$ Gaming: $\frac{6}{60} \times 360 = 36°$ Messaging: $\frac{14}{60} \times 360 = 84°$  	**Scale 10C (0, 3, 6, 10)** Allow a tolerance of $\pm 2°$ *Low Partial Credit:* • Work of merit, for example, $\frac{30}{60}$, $\frac{10}{60}$, $\frac{6}{60}$, or $\frac{14}{60}$; uses 360 • One angle correctly drawn or calculated *High Partial Credit:* • Angles correct and two angles correct in pie chart • Fully correct pie chart with no workings *Full Credit -1:* • Sectors not labelled and/or angles not indicated

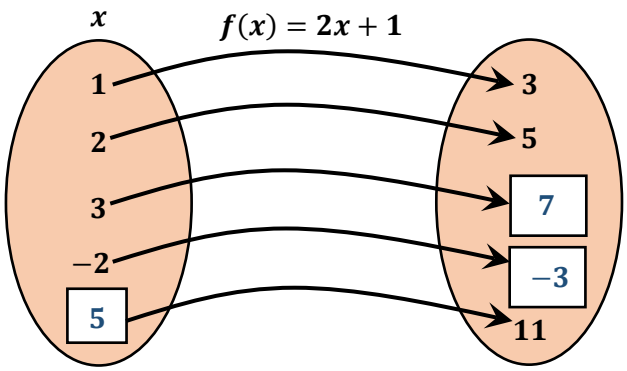
Q3	Model Solution – 25 marks	Marking Notes
(a)	**(i)** $21\,250 \div (0{\cdot}85) = [€]\,25\,000$ **OR** $25\,000 \times 0{\cdot}85 = [£]\,21\,250$ **OR** $\frac{21\,250}{25\,000} = 0{\cdot}85$ **(ii)** $\frac{3375}{25\,000} \times 100$ $= 13{\cdot}5\,\%$	**15D (0, 4, 8, 12, 15)** *Low Partial Credit:* • Work of merit in one part, for example, in **(i)**, mentions division of $0{\cdot}85$; mentions multiplication/division of $25\,000$; in **(ii)**, mentions division of $25\,000$; shows some knowledge of percentages *Mid Partial Credit:* • One part correct • Work of merit in both parts *High Partial Credit:* • One part correct and work of merit in the other part
(b) (i)(ii)	**(i)** Income at 20%: $40\,000 \times 0{\cdot}20 = [€]\,8000$ Income at 40%: $10\,000 \times 0{\cdot}40 = [€]\,4000$ **(ii)** Gross tax $= 8000 + 4000$ $= 12\,000$ Income tax $= 12\,000 - 3\,300$ $= 8700$ Net income $= 50\,000 - 8700$ $= [€]\,41\,300$ **OR** $50\,000 - 12\,000 + 3300 = [€]\,41\,300$	**Scale 10D (0, 3, 5, 7, 10)** Accept correct answers in part **(i)** without supporting work Consider the solution as consisting of 4 steps: **Step 1.** Calculates tax at 20% **Step 2.** Calculates tax at 40% **Step 3.** Handles tax credits correctly **Step 4.** Calculates net annual income *Low Partial Credit:* • Work of merit in one part, for example, in **(i)**, indicates $0{\cdot}2$ or $0{\cdot}4$ or calculates $10\,000$; shows some knowledge of percentages; in **(ii)**, relevant use of tax credits • 1 step correct *Mid Partial Credit:* • 2 steps correct *High Partial Credit:* • 3 steps correct

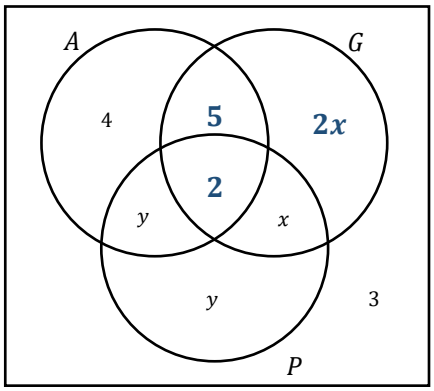
Q3	Model Solution – 25 marks	Marking Notes
(b) (iii)	**Method 1:** 60% of pay rise $= \ 4800$ 1% of pay rise $= \frac{4800}{60} = 80$ 100% of pay rise $= 80 \times 100$ $= 8\,000$ gross income $= 50\,000 + 8000$ $= [€]\ 58\,000$ **OR** **Method 2:** $41\,300 + 4800 = 46\,100$ $46\,100 - 3300 = 42\,800$ $42\,800 - 32\,000 = 10\,800$ $60\% = 10\,800$ $100\% = 10\,800 \times \frac{100}{60} = 18\,000$ $18\,000 + 40\,000 = [€]\ 58\,000$ **OR** **Method 3:** *New Gross Pay:* x *Tax payable:* $8000 + 0{\cdot}4(x - 40\,000) - 3300$ $0{\cdot}4\,x - 11\,300$ *Net Pay*: $41\,300 + 4800 = x - [0{\cdot}4\,x - 11\,300]$ $41\,300 + 4800 = 0{\cdot}6\,x + 11\,300$ $0{\cdot}6\,x = 34\,800$ $x = \frac{34\,800}{0{\cdot}6} = 58\,000$	**Scale 5C (0, 2, 3, 5)** *Low Partial Credit:* • Work of merit, for example, mentions 40% or 60%; $41\,300 + 4800$; some effort at finding tax payable in terms of x in method 3 *High Partial Credit:* • Calculates 8000 in method 1 • Calculates $18\,000$ in method 2 • Calculates $34\,800$ in method 3

Q4	Model Solution – 20 marks	Marking Notes
(a)	$B = (5,0) \quad C = (4,3)$	**Scale 5B (0, 3, 5)** *Partial Credit:* • Work of merit, for example, x or y co-ordinate correct; x and y swapped but otherwise correct for either point *Full Credit (-1):* • Interchanges B and C
(b)	$(2,-1) \in AC$ **OR** $(4,3) \in AC$ $(-1) = 2(2) + k$ $\quad$ $(3) = 2(4) + k$ $-4 - 1 = k$ $\quad$ $3 - 8 = k$ $-5 = k$ $\quad$ $-5 = k$ (or similar) **OR** $\overrightarrow{CA} = x: -2$ and $y: -4$ image of A under $\overrightarrow{CA} = (0,-5)$ $AC: y = 2x - 5$ $\Rightarrow k = -5$	**Scale 5C (0, 2, 3, 5)** Accept correct answer without supporting work *Low Partial Credit:* • Work of merit, for example, some correct substitution into the line; extension of AC; recognises slope is 2; substitution of slope into the equation of a line *High Partial Credit:* • Fully correct substitution into the line • $(0,-5)$ found by translation or graphing the line
(c)	*Answer:* True $\lvert AB\rvert = \sqrt{(5-2)^2 + (0--1)^2} = \sqrt{10}$ $\lvert BC\rvert = \sqrt{(4-5)^2 + (3-0)^2} = \sqrt{10}$ $\lvert AB\rvert = \lvert BC\rvert$	**Scale 5D (0, 2, 3, 4, 5)** *Low Partial Credit:* • Work of merit, for example, some correct substitution into the distance formula • Correct answer with no supporting work *Mid Partial Credit:* • One correct length calculated *High Partial Credit:* • Correct calculations with no answer or incorrect answer (2 equal sides or all 3 sides)

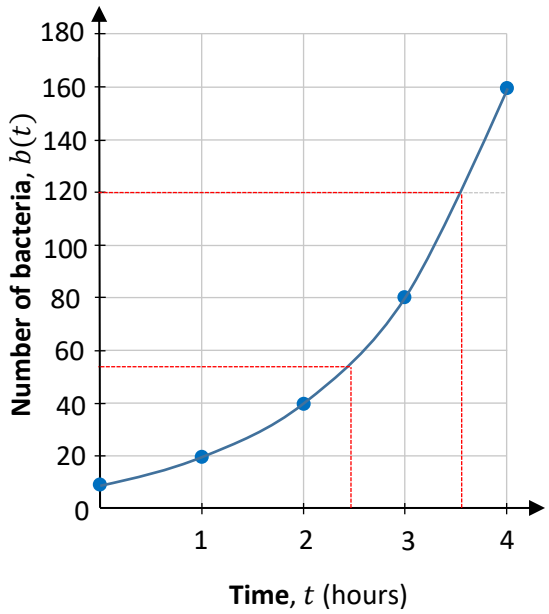
Q4	Model Solution – 20 marks	Marking Notes
(d)		**Scale 5C (0, 2, 3, 5)** *Low Partial Credit:* • Work of merit, for example, line drawn from A, B, or C to the origin; attempt at translation to origin; shows knowledge of central symmetry • One correct image point plotted *High Partial Credit:* • Two correct image points plotted *Full Credit (−1):* • Points not labelled

Q5	Model Solution – 15 marks	Marking Notes
(a)	$9 + 11 = 20$ $120 \div 20 = 6$ $9 \times 6 = [€]\ 54$	**Scale 5C (0, 2, 3, 5)** *Low Partial Credit:* • Work of merit, for example, identifies 20; relevant fraction written; correct ratio written; some relevant multiplication or division *High Partial Credit:* • Finds €6
(b)	Time $= \frac{250\ 000\ 000\ \text{km}}{300\ 000\ \text{km/sec}} = \frac{2500}{3}$ seconds $= 833{\cdot}\dot{3}$ seconds $\div 60$ $= \frac{125}{9}$ or $13{\cdot}8888\ldots = 13{\cdot}9$ [minutes] **OR** Speed: $300\ 000$ km/sec$= 18\ 000\ 000$ km/min Time$= \frac{250\ 000\ 000\ \text{km}}{18\ 000\ 000\ \text{km/min}} = 13{\cdot}9$ [minutes]	**Scale 10C (0, 3, 6, 10)** *Low Partial Credit:* • Work of merit, for example, 250 million correctly written as an integer; effort at changing units to minutes; correct relevant formula • Finds $18\ 000\ 000$ km/min *High Partial Credit:* • Answer given in seconds $(833{\cdot}3)$ *Full Credit – 1:* • No rounding or incorrect rounding

Q6	Model Solution – 10 marks	Marking Notes
(a)		**Scale 5C (0, 2, 3, 5)** Accept correct answer without supporting work *Low Partial Credit:* • Work of merit, for example, some correct substitution into $f(x)$ • One correct value *High Partial Credit:* • Two correct values
(b)	*Answer:* Diagram 2 *Reason:* Every input does not have an output **OR** One input has two outputs	**Scale 5B (0, 3, 5)** *Partial Credit:* • Correct diagram indicated with incorrect or no reason • Work of merit in the reason

Q7	Model Solution – 30 marks	Marking Notes
(a) (i)	These students solved the probability and geometry problems but not the algebra problem **OR** These students solved the probability and geometry problems **only**	**Scale 5B (0, 3, 5)** Note: Accept "$(G \cap P)\backslash A$" as correct *Partial Credit:* • Work of merit, for example, partially correct statement (eg. geometry and probability); $(G \cap P)$
(a) (ii)		**Scale 10C (0, 3, 6, 10)** Accept correct answers on diagram without supporting work *Low Partial Credit:* • Work of merit in calculating one entry *High Partial Credit:* • Two correct entries
(b) (i)	$\#(A \cup G) = 29$ $4 + 5 + 2 + 2x + x + y = 29$ $3x + y + 11 = 29$ $3x + y = 18$ $\#P = 13$ $2 + x + y + y = 13$ $2 + x + 2y = 13$ $x + 2y = 11$	**Scale 5D (0, 2, 3, 4, 5)** *Low Partial Credit:* • Work of merit in finding one equation *Mid Partial Credit:* • One correct equation found • Work of merit in finding both equations *High Partial Credit:* • One equation found and work of merit in finding the other *Full Credit (-1):* • Correct equations from their incorrect work but with no conclusion

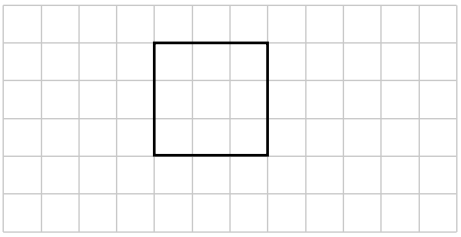
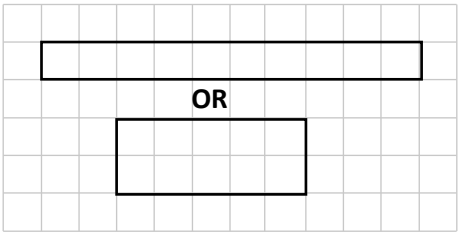
Q7	Model Solution – 30 marks	Marking Notes
(b) (ii)	$6x + 2y = 36$ $\underline{-x - 2y = -11}$ $5x = 25$ $x = 5$ $3(5) + y = 18$ $15 + y = 18$ $y = 3$ **OR** $3x + y = 18$ $\underline{-3x - 6y = -33}$ $-5y = -15$ $y = 3$ $3x + (3) = 18$ $3x = 15$ $x = 5$ **OR** $x = 11 - 2y$ $3(11 - 2y) + y = 18$ $33 - 6y + y = 18$ $5y = 15$ $y = 3$ $3x + (3) = 18$ $3x = 15$ $x = 5$	**Scale 10D (0, 3, 5, 7, 10)** Consider solution is consisting of 4 steps: **Step 1.** Multiplies equation(s) so that 1 variable will cancel / express one variable in terms of the other **Step 2.** Produces one equation in one variable **Step 3.** Finds value of one variable **Step 4.** Finds value of second variable *Low Partial Credit:* • Work of merit, for example, relevant work in isolating one variable in one equation or indicates multiplying one equation by constant *Mid Partial Credit:* • 2 steps correct *High Partial Credit:* • 3 steps correct

Q8	Model Solution – 25 marks	Marking Notes
(a)	**(i)** <table><tr><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>**10**</td><td>**20**</td><td>**40**</td><td>80</td><td>**160**</td></tr></table> **(ii)** 	**Scale 10D (0, 3, 5, 7, 10)** Note: Award LPC at most if function is treated as linear Consider solution as consisting of 10 parts: **4 parts:** 4 values in the table in part **(i)** **5 parts:** 5 pointed plotted from the values in the table **1 part:** All points joined appropriately (must be exponential) *Low Partial Credit:* • 1 part correct *Mid Partial Credit:* • 6 parts correct *High Partial Credit:* • 8 parts correct *Full Credit (–1):* • Fully correct apart from 1 part
(b)	**(i)** 57 [bacteria] **(ii)** 3·6 [hours]	**Scale 5D (0, 2, 3, 4, 5)** *Note*: One part is considered correct if it has supporting work on graph *Note*: Accept answers consistent with candidate's graph *Note*: Accept answers between 50 and 60 in **(i)** and between 3·3 and 3·8 in **(ii)** *Low Partial Credit:* • Work of merit in one part, for example, relevant work on diagram; one correct value given without supporting work on graph *Mid Partial Credit:* • One part correct • Work of merit in both parts *High Partial Credit:* • One part correct and work of merit in the other part

Q8	Model Solution – 25 marks	Marking Notes
(c)	*Answer:* Exponential *Reason:* It's doubling each time	**Scale 5B (0, 3, 5)** Note: Accept answer consistent with candidate's table *Partial Credit:* • Correct answer with incorrect or no reason • Incorrect answer with work of merit in reason • Shows understanding of linear/quadratic/exponential functions
(d)	$10(2^{24}) = 167\ 772\ 160$ $= \ 1{\cdot}67772160 \times 10^8$ or $168\ 000\ 000$ $= \ 1{\cdot}68 \times 10^8$	**Scale 5C (0, 2, 3, 5)** Accept correct answer without supporting work *Low Partial Credit:* • Work of merit, for example, correct substitution into function; $t = 24$ *High Partial Credit:* • Finds $167\ 772\ 160$ *Full Credit (-1):* • No rounding or incorrect rounding

Q9	Model Solution – 25 marks	Marking Notes
(a)	$x = \frac{-(7) \pm \sqrt{(7)^2 - 4(2)(-3)}}{2(2)}$ $= \frac{-7 \pm \sqrt{73}}{4}$ $x = 0{\cdot}386\ldots \quad x = -3{\cdot}886\ldots$ $x = 0{\cdot}39 \quad x = -3{\cdot}89$ [2 D.P.]	**Scale 5C (0, 2, 3, 5)** *Low Partial Credit* • Work of merit, for example, some substitution into correct formula • Identifies a or b or c • One correct answer without supporting work *High Partial Credit* • Full correct substitution into formula • Two correct answers without supporting work • One error in filling in formula, but evaluates correctly • $-7 \pm \frac{\sqrt{73}}{4}$ and finishes correctly *Full Credit −1* • No rounding or incorrect rounding
(b)	$\frac{3(x) - 4(x+1)}{x(x+1)}$ $= \frac{3x-4x-4}{x(x+1)}$ $= \frac{-x-4}{x(x+1)}$	**Scale 5C (0, 2, 3, 5)** *Low Partial Credit:* • Work of merit, for example, correct common denominator • $\pm[3(x) - 4(x+1)]$ and no denominator *High Partial Credit:* • $\frac{3(x) - 4(x+1)}{x(x+1)}$ • $\frac{3(x+1) - 4(x)}{x(x+1)}$ and finishes correctly • Shows correct common denominator in calculations but drops it subsequently • 2 terms correct in numerator and finishes correctly, assuming that the first line in the solution is not shown

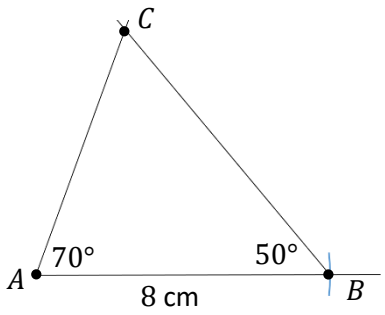
Q9	Model Solution – 25 marks	Marking Notes
(c)	Quadratic with roots -3 and -2: $(x+3)(x+2)$ $x^2+3x+2x+6$ x^2+5x+6 Cuts y-axis at $(0,12)$ $\therefore$ constant $=12$ $g(x)=2(x^2+5x+6)$ $g(x)=2x^2+10x+12$ $a=2, \quad b=10, \quad c=12$ **OR** $g(-3)=0: a(-3)^2+b(-3)+c=0$ $9a-3b+c=0$ $g(-2)=0: a(-2)^2+b(-2)+c=0$ $4a-2b+c=0$ $g(0)=20: a(0)^2+b(0)+c=12$ $c=12$ $9a-3b=-12$ $4a-2b=-12$ $18a-6b=-24$ $-12a+6b=36$ $6a=12$ $a=2$ $9(2)-3b=-12$ $-3b=-30$ $b=10$ $a=2, \quad b=10, \quad c=12$	**Scale 10D (0, 3, 5, 7, 10)** *Low Partial Credit:* • Work of merit, for example, one correct factor; some correct substitution into $g(x)$; finds c; sketch of quadratic graph; some understanding of roots *Mid Partial Credit:* • Finds $(x+3)(x+2)$ • $(-3,0)$ or $(-2,0)$ fully substituted into $g(x)$ *High Partial Credit:* • Finds x^2+5x+6 in the first solution • Finds $c=12$ and $a=2$ or $b=10$ in the second solution

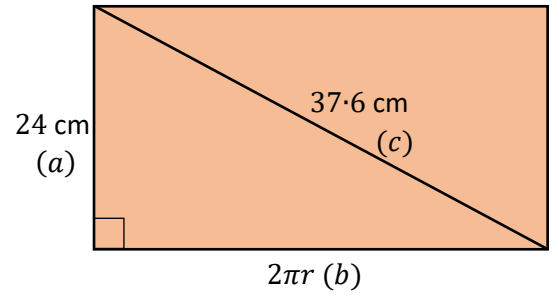
Q10	Model Solution – 25 marks	Marking Notes
(a)	**(i)**  **(ii)**  **OR**	**Scale 10C (0, 3, 6, 10)** *Low Partial Credit:* • Work of merit in one part, for example, one correct side drawn; some relevant multiplication or division *High Partial Credit:* • One part correct
(b)(i)	Area $= \frac{1}{2}\ (b)(h)$ $\frac{1}{2}(k)(h) = 4h$ $\frac{1}{2}k = 4 \qquad (\times 2)$ $k = 8$	**Scale 5C (0, 2, 3, 5)** *Low Partial Credit:* • Work of merit, for example, some correct substitution into formula or lets $\frac{1}{2}bh$ equal $4h$ *High Partial Credit:* • Finds $\frac{1}{2}k = 4$

Q10	Model Solution – 25 marks	Marking Notes
(b)(ii)	Volume $= 4h\,(h+4) = 4h^2 + 16h$ $4h^2 + 16h = 240$ $h^2 + 4h - 60 = 0$ $(h+10)\,(h-6) = 0$ $h = -10$ or $h = 6$	**Scale 10D (0, 3, 5, 7, 10)** Note: Accept correct answer obtained through trial and error if verified for full credit Consider solution as consisting of 4 steps: **Step 1.** Finds volume of prism in terms of h **Step 2.** Sets up quadratic equation **Step 3.** Fully substitutes quadratic formula or finds factors **Step 4.** Finds the value of h *Low Partial Credit:* • Work of merit in one part, for example, volume of prism = area of the cross section × length, or area of the triangle × length • $4h(h+4)$ *Mid Partial Credit:* • Two parts correct *High Partial Credit:* • Three parts correct *Full Credit (−1):* • Two values of h found as per the equation, but negative one not excluded/positive one not selected

Q11	Model Solution – 25 marks	Marking Notes
(a)	*Answer:* Yes *Calculations:* slope $= \frac{40}{600}$ or $\frac{1}{15}$ $\quad \left[\frac{1}{15} < \frac{1}{12}\right]$	**Scale 5C (0, 2, 3, 5)** *Low Partial Credit:* • Work of merit, for example, converts lengths to same unit; finds $\frac{40}{6}$ • Correct answer with no merit in calculations *High Partial Credit:* • Incorrect answer with correct calculations • Correct answer with work of merit in the calculations
(b)	$\lvert AC\rvert^2 = 40^2 + 600^2$ $= 361\,600$ $\lvert AC\rvert = \sqrt{361\,600}$ $= 601{\cdot}33 \ldots$ $= 601{\cdot}3$ [cm]	**Scale 5C(0, 2, 3, 5)** *Low Partial Credit:* • Work of merit, for example, some correct substitution into Pythagoras' theorem; effort at finding θ using tan *High Partial Credit:* • Finds $361\,600$ • Finds θ and then isolates AC using trigonometric ratios • Correct answer without supporting work *Full Credit (−1):* • No rounding or incorrect rounding

Q11	Model Solution – 25 marks	Marking Notes
(c)(d)	**(c)** $\sin\theta = \frac{40}{601\cdot3}$ $\theta = \sin^{-1}\left(\frac{40}{601\cdot3}\right)$ $\theta = 3\cdot8°$ **OR** $\tan\theta = \frac{40}{600}$ $\theta = \tan^{-1}\frac{40}{600}$ $\theta = 3\cdot8°$ **OR** $\cos\theta = \frac{600}{601\cdot3}$ $\theta = \cos^{-1}\left(\frac{600}{601\cdot3}\right)$ $\theta = 3\cdot8°$ **(d)** $\sin\theta = \frac{\text{Opp}}{\text{Hyp}}$ $\sin\theta = \frac{x}{7\cdot5}$ $x = 7\cdot5(\sin 2\cdot9)$ $x = 0\cdot379447\ldots.$ $x = 38$ [cm]	**Scale 10D (0, 3, 5, 7, 10)** Note: Accept correct answer without work if θ was found in the previous part *Low Partial Credit:* • Work of merit in one part, for example, relevant trigonometric ratio • Correct labelling of 2 relevant sides in triangles in **(c)** and/or **(d)** *Mid Partial Credit:* • One part correct • Work of merit in both parts *High Partial Credit:* • One part correct and work of merit in the other part *Full Credit (−1):* • No rounding or incorrect rounding • Calculator in incorrect mode

Q12	Model Solution – 20 marks	Marking Notes
(a)	$a = 96 + 20$ $= 116°$ **OR** 3rd angle: $180 - (96 + 20) = 64$ $a = 180 - 64 = 116°$ $b = \frac{1}{2}(96) = 48°$	**Scale 10C (0, 3, 6, 10)** Accept correct answers without work *Low Partial Credit:* • Work of merit, for example, the sum of the angles in a triangle is $180°$; the measure of the angle at the centre is twice the measure of the angle at a point on the circle *High Partial Credit:* • Finds a or b
(b)		**Scale 10C (0, 3, 6, 10)** Note: Allow a tolerance of $\pm 2°$ and ± 2mm *Low Partial Credit:* • Work of merit for example, relevant arc or angle drawn • Ray drawn from A to B *High Partial Credit:* • Any two of the following drawn: $[AB]$, $\lvert\angle CAB\rvert$ or $\lvert\angle ABC\rvert$ *Full Credit (−1):* • B and/or C not labelled

Q13	Model Solution – 15 marks	Marking Notes
	 $a^2 + b^2 = c^2$ $24^2 + b^2 = 37{\cdot}6^2$ $b = \sqrt{37{\cdot}6^2 - 24^2}$ $b = 28{\cdot}94408402$ $2\pi r = 28{\cdot}94408402$ $r = \frac{28{\cdot}94408402}{2\pi}$ $r = 4{\cdot}6$ [cm]	**Scale 15D (0, 4, 8, 12, 15)** *Low Partial Credit:* • Work of merit, for example, some correct substitution into Pythagoras' Theorem; mentions $2\pi r$; identifies width of rectangle as 24; links length of circle to length of rectangle; links $2\pi rh$ to area of rectangle *Mid Partial Credit:* • Finds $28{\cdot}944$ … *High Partial Credit:* • $2\pi r = 28{\cdot}944$ … *Full Credit (−1):* • No rounding or incorrect rounding